

ALEJANDRO LOPEZ RUIZ

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PROFILE

Data Science and Artificial Intelligence student focused on applied analytics, machine learning, and data-driven decision support. Experienced with building end-to-end data products, including deployed NLP dashboards, client-facing web applications, data visualization systems, and deep learning models. Strong foundation in Python, SQL, statistics, ML, and software engineering. With interest in business analytics, AI, and sustainability-oriented projects.

EDUCATION

- Leiden University - BSc Data Science and Artificial Intelligence** Expected Jul 2026
- Relevant coursework: Machine Learning, Deep Learning, Databases, Statistics, Cybersecurity, Algorithms, Reinforcement Learning, Computer Vision, Natural Language Processing and Data Visualization.
 - Minor in Economics: Microeconomics, Macroeconomics, Finance, and Applied Data Science and Econometrics projects.
- Agora Lledo International School - International Baccalaureate, Bilingual Diploma** 2021 - 2023

SELECTED PROJECTS

- Marketing Social Media Analyzer - Solo NLP and Analytics Dashboard** Completed / Deployed
- Built a multi-language social listening dashboard that collects posts by keyword, detects language, translates them, and runs sentiment, emotion, and toxicity analysis for marketing-style insights.
 - Designed the full workflow from data collection to BI-style Streamlit visualization, demo available through my personal portfolio.
 - Tech: Python, Streamlit, NLP, Hugging Face models, sentiment analysis, data visualization, dashboard design.
- PEPAdb Archaeology Database Webpage - Client Data Visualization Platform** In progress / Ends Jun 2026
- Developing a professional end-to-end web application for the Archaeology Department of the University of Sevilla as part of a Software Engineering client project.
 - Main contribution: co-designed frontend architecture, data visualization, record filtering, visual design, and ML prediction; the team follows a professional workflow using Scrum, GitHub Actions, CI/CD, and code review practices.
 - Tech: React, Vite, FastAPI, REST APIs, Docker, GitHub Actions, CI/CD, Databases, Machine Learning, dashboard design.
- RouteGuesser - Software Development Project** Sep 2024 - Jan 2025
- Collaborated in a 6-person team to design and deploy a map-based game using Leiden city data and shortest-path algorithms; contributed to frontend, backend integration, and project coordination; accessible through my personal portfolio.
 - Final grade: 8.5; code quality: 10, design and quality: 9.5.
 - Tech: HTML, CSS, JavaScript, Python, Flask, Docker, Google Cloud Run.
- EcoSim RL Ecosystem Simulator - Bachelor Thesis Project** In development
- Building a reinforcement-learning ecosystem simulation to study predator-prey dynamics as my thesis project. Designed to explore how agent-based RL simulations can inform sustainability-oriented ecological policy and biodiversity conservation.
 - Tech: Python, agent-based modeling, Reinforcement Learning, environment design and simulation, data analysis and evaluation.

WORK EXPERIENCE

- McDonald's - Crew Trainer** Apr 2024 - Present
- Promoted to Crew Trainer in Oct 2024 after demonstrating reliability, teamwork, and operational responsibility.
 - Train and mentor new staff while supporting team coordination, guest experience, food safety, hygiene standards, and rule compliance in a high-pressure and fast-paced service environment.

RELEVANT ACADEMIC HIGHLIGHTS

- Chest X-ray CNN and XAI project: manually tuned model reached 94% accuracy and 0.96 recall on the pneumonia class.
- Dashboard assignment graded 9.8; convolutional neural network for video analysis project graded 9.3.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, JavaScript, HTML/CSS, Pandas, NumPy, Jupyter, Stata.
Machine Learning & AI: PyTorch, supervised/unsupervised learning, model evaluation, HPO, CV, NLP, transformers, Hugging Face.
Visualization & Analytics: Matplotlib, Seaborn, Streamlit, dashboards, data storytelling, regression analysis, econometrics.
Software & Tools: Git, Docker, Linux, REST APIs, Flask, FastAPI, React, Vite, Excel, Google Cloud Run.

LANGUAGES

English: professional working proficiency, C1 | Spanish: Native proficiency | Dutch: Beginner